

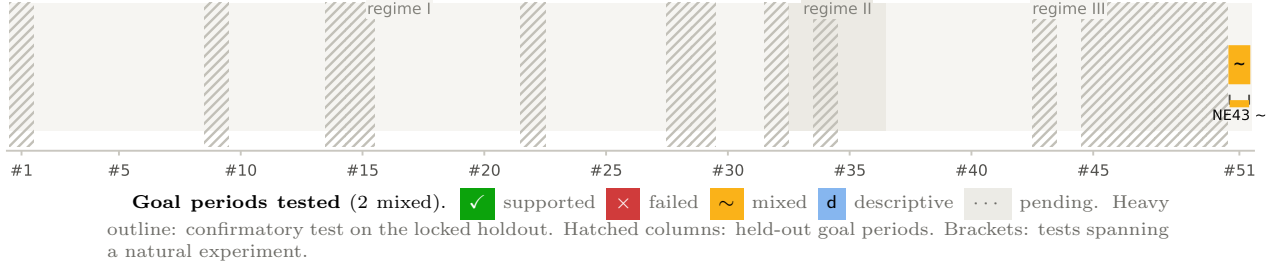
H09: Agent swarms have an effective thermodynamics

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1.5/5 Usefulness 2.5/5

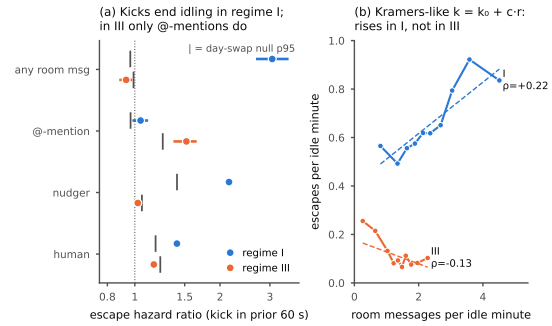
Bottom line. The thermodynamic framing adds little; the timer gate reads context (chatter holds, an @-mention releases), and memory size is a homeostat that overshoots after forced erasures.

level: hypothesis · pipeline checks 29% · updated 2026-10-04



WHAT HAPPENED

H09 is an exploratory framing, not one test. It asks whether swarm statistics read as thermodynamics: landscapes, currents, traps and equations of state. Two light non-holdout rounds and round 1b exist; no holdout has run. In round 1 the daily schedule, a field, reproduces the two wells of the activity landscape. Per-agent action chains are irreversible: 78% of cells beat the null. Regime III has a closed cycle consolidate → search → talk → pause. Event-level entropy production collapses after perma-computer-use. Agents are ergodic at day scale, and restarts relax within a minute. In regime I, room messages raise the escape from idle spells (hazard ratio 3.06; null p95 0.96). In regime III a pause is a self-set timer (0.22% end early). Round 1b reads the gate on the context ledger: unaddressed chatter keeps agents paused (OR 0.50), and an @-mention releases them (OR 3.1). Memory size is an agent-specific set point that overshoots after forced erasures. Collective entropy production is small (plug-in only).



Observables. (a) The escape hazard ratio of idle spells: minutes with a kick in the previous 60 s against minutes without, stratified by elapsed dwell and time of day. Bars: agent-day bootstrap CIs where computed; ticks: 95th percentile of a day-swap null. In regime I any room message triples escape. In regime III room messages do not help; only @-mentions do. (b) The escape rate against the kick rate over agent-day deciles (dashed: linear fit $k_0 + cr$). It rises in I and falls in III.

PHYSICAL MODEL

No single model: each sub-claim borrows one (models 01, 02, 09, 10). $\lambda(t)$ is the escape hazard of an idle spell at elapsed dwell t . HR is the hazard ratio for a kick in the prior 60 s against none. k_0 and c are the base escape rate and the kick sensitivity, and r is the kick rate. t^* is the declared pause expiry, and p_{gate} the chance of acting at expiry. π and P are the stationary law and transitions of an agent's action-class chain. V is the memory size, P_{in} the inflow, β the response and γ the relaxation.

$$\lambda(t) = P(\text{escape at } t \mid \text{idle until } t), \quad \text{HR} = \lambda_{\text{kick}} / \lambda_{\text{none}}$$

regime I (Kramers-like): $k \approx k_0 + cr$

regime III (timer gate): $\lambda(t) \approx \delta(t - t^*) p_{\text{gate}} (\text{@-mention})$

$$\sigma = \sum_{x < y} (\pi_x P_{xy} - \pi_y P_{yx}) \ln \frac{\pi_x P_{xy}}{\pi_y P_{yx}}$$

$$\Delta V = \beta P_{\text{in}} - \gamma V_{\text{prev}} \Rightarrow V^* = \beta P_{\text{in}} / \gamma$$

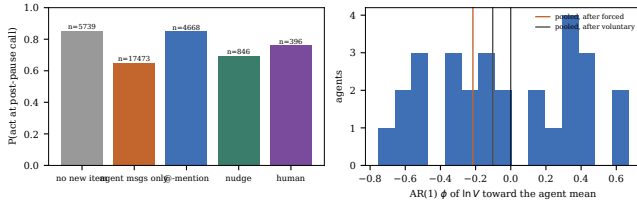
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

The thermodynamic vocabulary does little work; the useful findings are plain mechanics. In the current scaffold (regime III) an idle agent sleeps on a timer that it set itself. Unaddressed room chatter keeps it asleep: it acts at 65% of gates, against 85% with no news. To wake a timer-paused agent, address it by name, and expect a delay of up to its chosen pause. Regime-I agents were interruptible. Memory size is a set point: consolidation resets it, and storage does not grow. The rest restates the scaffold: schedule-made landscapes, day-scale ergodicity, and a consolidation cycle that is the prompt's own loop. Nothing here yet beats a simpler statistic.

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RESULTS VS PREDICTIONS

Prediction	Round 1/2 → round 1b	Verdict (1b)
E1 VR > 1.5 (regime III)	1.61 → 1.85; trimmed to day edges 1.22	no (day edges)
E6c @-mention HR (co- incidence)	1.52 → identical (clean men- tions)	yes
G1a @-mention at the ledger gate	new: OR 3.1 [1.9, 4.5]	yes
G1b unaddressed chatter neutral	new: OR 0.50 vs no news (0.65 vs 0.85 act)	no
N1a gate stable at NE43 steps	log ratio -0.01, -0.05	yes
N1b nudge accounting	drop 0.036 > bound 0.013	no
E8a elasticity on inflow	0.16 (exposure) → 0.14 (ledger reads)	clock, inflow not
N2a #51 V vs reads per call	0.02 [-0.02, 0.06]	yes
N2b #51 V vs N	0.33 [0.03, 0.64]	no (tenure?)
N3 homeostat, no overshoot	ϕ median -0.10; after forced -0.22	overshoot (HH269)



Second figure. (a) Regime III, non-holdout: the probability that an agent acts, rather than re-pauses, at the call after a timer pause, by what newly enters that call in the context ledger. (b) The per-agent AR(1) coefficient of log memory size toward the agent's mean. Lines: pooled coefficient after forced (41-turn cap) and voluntary consolidations.

GOAL PERIODS IN DETAIL

G51 ~ 1b: mixed (V ignores read inflow but rises with N; th...
 NE43 ~ 1b: mixed (the gate is unchanged; overall acting fell...

FAITHFULNESS SCORECARD

A mapping	–	F identifiability	–
B assumptions	–	G ground truth	–
C adequacy	–	H comparative	–
D unfitted predictions	–	I transfer	–
E interventional	–		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The gate contrasts are within-agent but not randomized: chatter arrives when other agents are busy. In #51, time confounds N2's N effect. E7 (pair EP) did not re-run on fixed bins. No scorecard axis has a synthetic check.

ROUND 2

Fit a two-parameter gate model per agent: chatter suppression and mention release. Test HH269's gain at NE14 with the ledger's reset types.

- **Essence.** Tokens are the energy, progress is the work, and context resets are the heat.
- **H09-R1** Token thermodynamics: power = tokens per hour; efficiency = progress (work ledger) per token; loops waste tokens at zero efficiency.
- **H09-R2** Consolidation is Landauer-like erasure: tokens spent consolidating vs information kept.
- **H09-R3** The swarm is a driven-dissipative steady state whose throughput is limited by attention, not compute (E5).

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